AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 4 of 9

21 marks · about 25 minutes

Key information

- Exothermic forward reaction: raising temperature lowers K; endothermic: raising temperature increases K
- Units: substitute the units into the expression and cancel (e.g. kPa^{-2}, $mol^{-1}\ dm^3$ or no units)
- For the reverse reaction K' = 1/K; halving the equation gives √K
- Mole fraction = moles of that gas ÷ total moles of gas
- Partial pressure = mole fraction × total pressure

Section A: Short-answer questions

01 Give **two** features of a reaction in dynamic equilibrium.

[2 marks]

02 For $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, state the effect of increasing the total pressure at constant temperature on the equilibrium yield of PCl_3 and on the value of K_p.

[2 marks]

03 Give an expression for K_c for $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ and deduce its units.

[2 marks]

04 Nitrogen dioxide can be produced in a reaction between ammonia and oxygen.

$4NH_3(g) + 7O_2(g) \rightleftharpoons 4NO_2(g) + 6H_2O(g) \quad \Delta H^{\ominus} = -1132\ kJ\ mol^{-1}$

Some ammonia and oxygen are reacted and the mixture is left to reach equilibrium at a fixed temperature. Explain why increasing the temperature of this equilibrium mixture decreases the mole fraction of NO_2 in the equilibrium mixture.

[2 marks]

Section B: Longer questions

05 Carbon monoxide and hydrogen react to form methanol.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

05.1 0.430 mol of carbon monoxide is mixed with 0.860 mol of hydrogen. At equilibrium, the total pressure in the flask is 250 kPa and the mixture contains 0.110 mol of methanol. Calculate the amount, in moles, of carbon monoxide present at equilibrium. Calculate the partial pressure, in kPa, of carbon monoxide in this equilibrium mixture.

[3 marks]

05.2 Give an expression for the equilibrium constant (K_p) for this reaction.

[1 mark]

05.3 A different mixture of carbon monoxide and hydrogen is left to reach equilibrium at a temperature T.

	Value
Partial pressure of CO	125 kPa
Partial pressure of CH_3OH	5.45 kPa
K_p	$1.15 \times 10^{-6}\ kPa^{-2}$

Calculate the partial pressure, in kPa, of hydrogen in this equilibrium mixture.

[3 marks]

05.4 Use the K_p value from the table to calculate a value for K_p for the reaction $CH_3OH(g) \rightleftharpoons CO(g) + 2H_2(g)$ at temperature T. Give the units for K_p.

[2 marks]

06 This reaction can be used to make hydrogen.

$H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g) \quad \Delta H = -41 \text{ kJ mol}^{-1}$

06.1 A mixture of 2.00 mol of $H_2O(g)$ and 2.00 mol of $CO(g)$ is allowed to reach equilibrium at a constant temperature in a 20 dm^3 container. At equilibrium, there are 0.92 mol of $H_2(g)$. Calculate the mole fraction of $H_2(g)$ in the equilibrium mixture.

[2 marks]

06.2 State why the equilibrium constant (K_p) for this reaction has no units.

[1 mark]

06.3 The temperature of the equilibrium mixture is increased. State how the amount of $H_2(g)$ changes when the new position of equilibrium is reached.

[1 mark]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	forward and reverse reactions proceed at equal rates concentrations (of reactants and products) remain constant	*Source: 7405/1 Jun 2022 Q01.1* Do not accept equal concentrations; ignore closed system	2
02	yield of PCl_3 decreases K_p no change	*Source: Practice question (written for this set)*	2
03	$K_c = [NH_3]^2 \div [N_2][H_2]^3$ $mol^{-2}\ dm^6$	*Source: Practice question (written for this set)*	2
04	M1 (forward) reaction is exothermic M2 equilibrium shifts to lower the temperature / to oppose the increase in temperature	*Source: 7405/1 Jun 2025 Q04.5* M1 allow reverse reaction is endothermic Ignore favours; 'shifts to the left' alone is insufficient for M2	2
05.1	M1 n(CO) = 0.32 mol M2 total moles = 1.07 mol / mole fraction CO = 0.299 M3 p(CO) = 0.320 × 250 ÷ 1.07 = 74.8 kPa	*Source: 7405/1 Jun 2022 Q01.3–01.6* Allow 75 kPa 67.8 kPa = 2 marks max	3
05.2	$K_p = p(CH_3OH) \div (p(H_2)^2 \times p(CO))$	Do not accept square brackets	1
05.3	M1 $p(H_2)^2 = p(CH_3OH) \div (K_p \times p(CO)) = 5.45 \div (1.15 \times 10^{-6} \times 125)$ M2 $p(H_2)^2 = 37\ 913$ M3 $p(H_2) = 194.7$ kPa	Allow 195 kPa If rearrangement incorrect allow M3 only	3
05.4	$1 \div 1.15 \times 10^{-6} = 8.7(0) \times 10^5$ kPa^2		2
06.1	M1 total amount of gas at equilibrium = 4.00 mol M2 mole fraction of H_2 = 0.23	*Source: 7405/1 Jun 2024 Q04.1–04.3* Accept fractions	2
06.2	(The amount in) moles of products is the same as reactants (so the units of partial pressure cancel)		1
06.3	The amount decreases	Original is tick-one-box (decreases / does not change / increases)	1